li.yusen.astr@gmail.com‡
+61 0401873569‡
+86 13691120947

李宇森 Li, Yusen

Education

University of Groningen Groningen, The Netherlands
PhD in Astronomy and Astrophysics 2026.09 - 2030.09
- **Supervisors:** Prof. Else Starkenburg, Prof. Amina Helmi

Australian National University Canberra, Australia
MSc (Adv) in Astronomy and Astrophysics 2025.07 - 2026.06
- **Supervisors:** Prof. Kenneth Freeman, Prof. Helmut Jerjen
- **Thesis:** The bar's influence on the diffuse ionised gas emission and motion in NGC 5746

BSc (Hons) in Astronomy and Astrophysics 2024.02 - 2024.12
- **Supervisors:** Prof. Kenneth Freeman, Prof. Helmut Jerjen
- **Thesis:** Orbital dynamics in barred Milky Way: the dynamical and chemical origin of the Hercules stream

Bachelor of Science (BSc) 2021.02 - 2023.12
- Double major in Mathematics & Astronomy and Astrophysics

Awards and Honors

Bok Prize (**Astronomical Society of Australia**) 2025.07
Outstanding Honours/Masters thesis in Australian Astronomy in 2024

Academic Experience

Vacation Studentship | CAS, Swinburne 2026.01 - 2026.02
Advisor: Dr. Rebecca Davies

Visitor | National Astronomical Observatories, China 2024.12 - 2025.06
Advisor: Prof. Jingkun Zhao

Publications (ADS library)

Li, Y., Freeman, K., & Jerjen, H. 2025a, Monthly Notices of the Royal Astronomical Society, 539, On the origin of the Hercules group: II. The Trojan quasi-periodic identity on the orbital level, 1595 (ADS entry)

Li, Y., Freeman, K., Jerjen, H., Buder, S., Hayden, M., & Mondal, A. 2025b, Monthly Notices of the Royal Astronomical Society, 538, On the origin of the Hercules group - I. Chemical signatures indicating the outer bar origin, 1963 (ADS entry)

Research Interest

Milky Way: Galactic archaeology, Galaxy structures and dynamics

Galaxies: Galactic dynamics, bars and spirals, formation and evolution, morphology and structures

Computing: Simulation, mathematical modelling, machine learning, `julia`

Conferences

The bar's influence on the ionised gas emission and motion in edge-on galaxies: NGC 5746 as a start Poster

The 2026 Astronomical Society of Australia Annual Scientific Meeting

On the chemical and orbital origin of Hercules Prize Talk

The 2025 Astronomical Society of Australia Annual Scientific Meeting

Stellar Orbit Dynamics Presentation

Mount Stromlo Student Seminar 2024

The dynamical origin of the Hercules stream Sparkler

The 2024 Astronomical Society of Australia Annual Scientific Meeting